Abdur Rahim

erahim3@gmail.com | arahim.dev | github.com/arahim3

Open Source

DigUp: Search your Computer by what's in your files GitHub (**6k**+ Stars)

- Built DigUp, an open-source, on-device semantic file search app that finds photos, PDFs, documents, audio, video and code by their content, using Google DeepMind's EmbeddingGemma 2 on llama.cpp.
- Shipped cross-platform on macOS, Windows and Linux from one shared Swift search engine, with a native SwiftUI app on Mac and a Tauri app on Windows and Linux, plus GPU acceleration through Metal and Vulkan.

mlx-tune: LLM/Multimodal Fine-Tuning Framework for Apple Silicon GitHub (**1.5k**+ Stars)

- Built and maintain an open-source MLX-native framework with an Unsloth-compatible API for LLM/VLM/Audio/OCR
- Supports LLMs, vision-language, speech (TTS/STT), embeddings, and OCR models via LoRA/QLoRA across 7 trainer algorithms (SFT, DPO, ORPO, GRPO, KTO, SimPO, CPT)
- Authored pure-MLX CTC/RNN-T/TDT transducer losses; first streaming-ASR training path on Apple Silicon, with LoRA support for 39+ MoE architectures and 50+ end-to-end examples

mlx-dspark: DeepSeek's DSpark Speculative decoding on Apple Silicon GitHub (**720**+ Stars)

- The MLX port of DeepSeek's DSpark speculative-decoding drafter: lossless **4x** faster Gemma-4/Qwen3/Qwen3.6 inference on Apple Silicon, served via an OpenAI-compatible API with continuous batching and prefix caching.
- Profiled and optimized the speculative-decoding path (cross-attention KV cache, Markov/confidence heads, 4-bit quantization); derived a memory-bandwidth cost model + DFlash/EAGLE3 benchmarks, and fixed a GQA KV-tiling bottleneck, turning a net-negative long-context regression into a flat up to 3× speedup at 12k+ tokens.

kaggle-tpu-lab: open-source LLM inference on free TPUs GitHub (**740**+ Stars)

- Run frontier-class open LLMs on Kaggle's free TPU v5e-8 as public OpenAI-/Anthropic-compatible endpoints
- GLM-5.3-Flash (320B MoE): first jax/pallas inference engine to run this model on TPU, 165 tok/s decode (262k ctx)
- Qwen3.8-27B (bf16, no quant, native 262k ctx): 130 tok/s single-stream decode and 10,300 tok/s prefill

Professional Experience

Senior AI Engineer ESAP AI, Saudi Arabia (Remote) May 2025 – Present

- Shipped production voice agents on Pipecat with multi-agent orchestration for enterprise ERP workflows
- Fine-tuned custom Ultravox speech-LLM and Meta's Omnilingual ASR for bilingual Arabic/English ASR
- Designed multi-model A100 inference stack achieving <500ms TTFB with custom streaming services over WebRTC
- Built LLM service layer integrating Qwen/Llama with RAG retrieval and dynamic tool selection for ERP-domain agents

Technical Skills

Engineering:	Python, Swift, SQL, Bash; PyPI packaging, GitHub Actions CI, Docker, Linux, AWS
ML & Training:	PyTorch, JAX/Pallas, MLX, Transformers, PEFT, TRL, LoRA/QLoRA, SFT/DPO/GRPO
Inference:	vLLM, llama.cpp/GGUF, speculative decoding, continuous batching, quantization, CUDA, TPU
Voice & Agents:	Pipecat, LiveKit, WebRTC, Whisper, Parakeet, Ultravox, Kokoro, Deepgram, LangGraph, MCP

Competitions

Kaggle Competitions Expert kaggle.com/rahim3 Dec 2019 – Present

- Google Landmark Recognition **Silver Medal** | Google Landmark Retrieval **Silver Medal**
- Biohub - Cell Tracking During Development, **Bronze Medal**

Education

University of Chittagong Chittagong, Bangladesh Jan 2017 – May 2023
Bachelor of Science in Electrical and Electronic Engineering